

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9648/02

Paper 2 Core Paper

2 September 2015

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.
Circle the question number of the question attempted.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
8	
Section B	
9 / 10	
Total	

This document consists of **22** printed pages and **1** blank page.

[Turn over

BLANK PAGE

Section A

Answer **all** the questions in this section.

- 1 Fig. 1.1 shows replication of a part of the glucagon receptor gene.

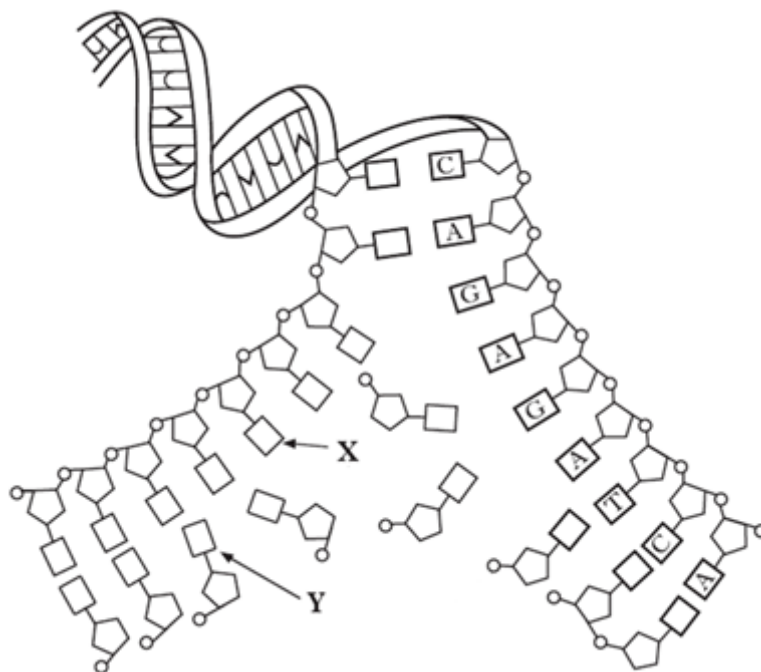


Fig. 1.1

- (a) (i) Name the bases labelled X and Y on Fig. 1.1. [1]

X: Cytosine;

Y: Thymine;

- (ii) State the phase of the cell cycle at which DNA replication occurs. [1]

1. S phase of interphase;;

- (iii) DNA replication involves a number of enzymes including DNA polymerase. Identify two other enzymes involved in DNA replication. [2]

1. helicase;;

2. DNA ligase;;

3. primase;;

4. DNA topoisomerase;;

(Reject: DNA polymerase)

- (iv) Explain how Fig. 1.1 shows semi-conservative replication. [2]

- 1. The two parental DNA strands separate due to the breaking of hydrogen bonds between complementary bases;;**
- 2. Both strands act as templates for the synthesis of new complementary DNA strands;;**
- 3. each new DNA molecule contains one original DNA strand and one newly synthesised DNA strand;;**

The structure of a glucagon receptor is shown in Fig. 1.2.

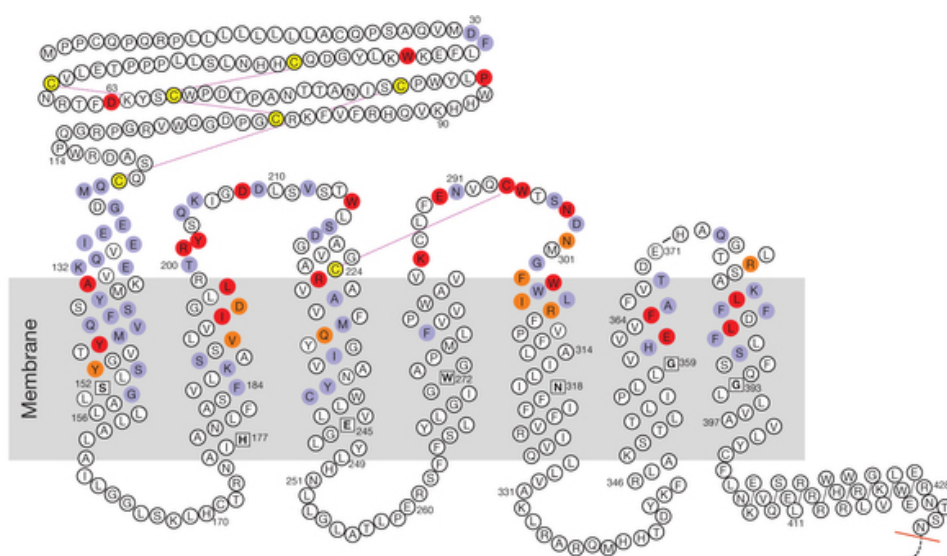


Fig. 1.2

(b) Describe how amino acid residues at different positions in the polypeptide may be brought together in the receptor site when the glucagon receptor is synthesized. [4]

1. Each protein contains a precise (number, type and) sequence of amino acids that are held together by peptide bonds (in a linear strand of polypeptide chain);;
2. The polypeptide chain will coil and fold into α -helix and β pleated sheet respectively (geometrically regular secondary structures as a result of hydrogen bonding between C=O and -NH groups of the amino acids);;
3. The polypeptide chain will undergo further bending, coiling, folding, to form the tertiary structure, (giving rise to a globular protein);;
4. Tertiary structure is held by ionic bonds, hydrogen bonds, disulphide bonds and hydrophobic interactions (any 3), formed between R groups of amino acids;;
5. All these levels of folding would enable the amino acid residues at different positions in the polypeptide to be brought together in the receptor site when the glucagon receptor is synthesized

In 2-8% of white populations, a mutation was incorporated in exon 2 of the glucagon receptor gene, which changes a glycine to a serine resulting in a Gly40Ser mutant receptor. The Gly40Ser mutant receptor was found to bind glucagon with an approximately three-fold lower affinity compared with the wild type receptor. This mutation located in the extracellular region of the glucagon receptor decreases the sensitivity of target tissues to glucagon.

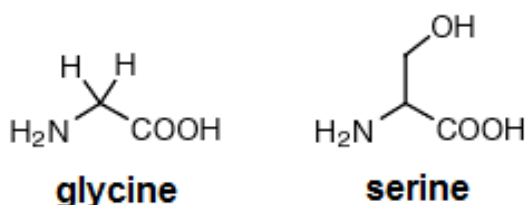


Fig. 1.3

(c) With reference to the information provided and Fig. 1.3, describe the effects of the mutation of the glucagon receptor gene. [4]

1. **A single base-pair substitution of the glucagon receptor gene causes (a) the DNA triplet to be changed in the gene/DNA / (b) one of the codons to be changed in the mRNA formed;;**
2. **Resulted in non-polar glycine being substituted by polar serine (at position 40) of the glucagon receptor (polypeptide) chain;;**
3. **This substitution changes the 3D conformation of the receptor / binding site resulting in glucagon unable to recognise and bind to the receptor / binding site, decreasing the affinity of glucagon to the receptor;; (Less cAMP activated, less protein kinase A activated, less phosphorylase kinase activated)**
4. **Less glycogen phosphorylase activated, decrease breakdown of glycogen to glucose /glycogenolysis;;**

[Total: 14]

- 2 Fig. 2.1 shows the main structural features of a mature (infective) human immunodeficiency virus (HIV).

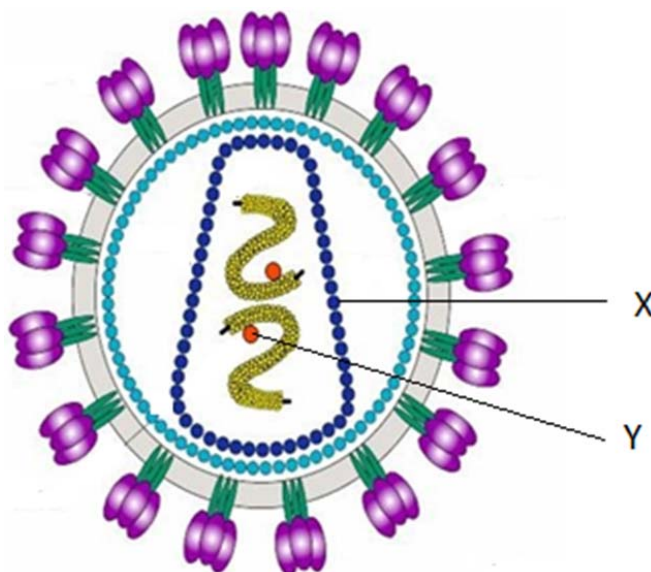


Fig. 2.1

(a) With reference to Fig. 2.1, label molecules X and Y. [2]

1. **X – capsomere / capsid protein;;**
2. **Y – reverse transcriptase / integrase;;**

(b) Describe the roles of HIV RNA in the infection process of T cells. [2]

HIV RNA codes for:

1. **reverse transcriptase which catalyses the synthesis of a DNA strand (complementary to HIV RNA) / cDNA;;**
2. **integrase which incorporate the double stranded DNA into the host cell's DNA (as a provirus);;**
3. **HIV protease which cleaves the immature HIV polyprotein into individual functional protein e.g. capsid, integrase molecules and reverse transcriptase molecules (1 eg);;**
4. **capsid which protects the nucleic acid from digestion by enzymes;;**
5. **gp120 / gp41 that aids in the entry of HIV into host cell;;**

(c) Raltegravir has been the first and is currently the only integrase strand transfer inhibitor (INSTI) approved for the treatment of HIV. Raltegravir inhibits integrase by binding to the active site and sequestering essential active site magnesium ions and the HIV genes are not expressed into HIV enzymes.

Suggest how Raltegravir work in the treatment of HIV infection. [2]

1. **Block the active site of integrase;;**
2. **double stranded DNA is unable to recognize and bind to active site;;**
3. **double stranded DNA will not be incorporated into the host cell's DNA as a provirus (by integrase);;**

Fig. 2.2 shows the structure of Middle East respiratory syndrome coronavirus (MERS-CoV). MERS affects the respiratory system (lungs and breathing tubes). Most MERS patients developed severe acute respiratory illness with symptoms of fever, cough and shortness of breath. About 3-4 out of every 10 patients reported with MERS have died.

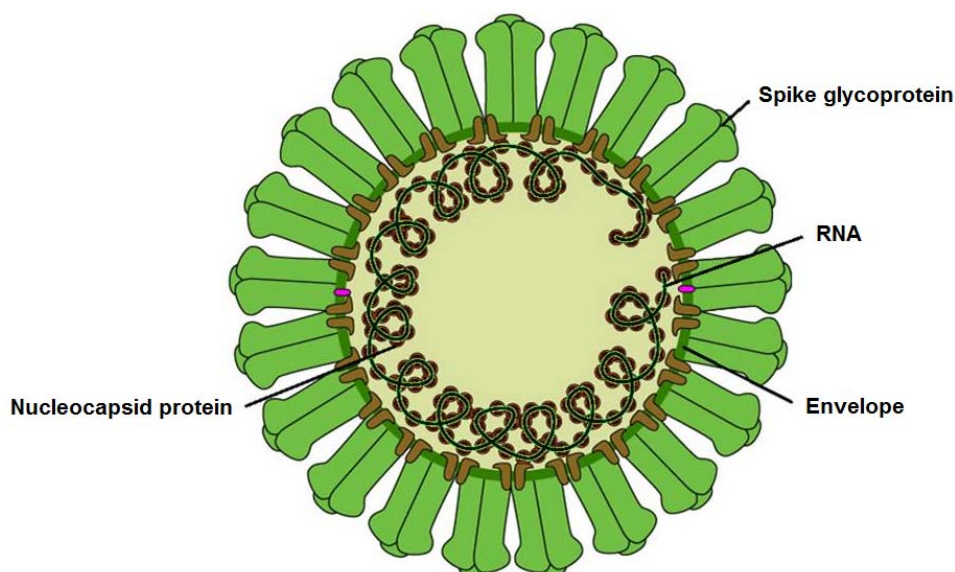


Fig. 2.2

(d) (i) With reference to Fig. 2.1 and Fig. 2.2, compare the structure of HIV and MERS-CoV. [2]

1. **Presence of RNA genome / (glyco)proteins / nucleocapsid proteins / envelope;;**
 2. **HIV has two copies of RNA while MERS-CoV has one copy;;**
 3. **Presence of capsid in HIV but absence of capsid in MERS-CoV;;**
 4. **Presence of enzymes within HIV but absence of enzymes within MERS-CoV;;**
- (pt 2-4: max 1)

(ii) The evidence available to date suggests that the coronavirus have been present in bats for some time and had spread to camels by the mid-1990s. The viruses appear to have spread from camels to humans in the early 2010s.

What changes may have occurred to give MERS-CoV the ability to infect humans? [2]

1. **(Point) mutations to spike glycoprotein gene in MERS-CoV during the replication of viral RNA (and accumulation of mutations result in gradual changes to MERS-CoV gene leading to antigenic drift);;**
2. **These cause a change in shape of the spike glycoprotein in MERS-CoV resulting in a better fit to specific binding sites of specific receptors on cell membrane of human host cells;;**

[Total: 10]

- 3 A recent study investigated a chromosome mutation discovered in malignant lung tumour cells. The mutation results in the fusion of the two genes shown in Fig. 3.1. The genes are *EML4*, which is involved in microtubule formation, and *ALK* which codes for a kinase enzyme. As a result of this fusion, a new protein (EML4-ALK) is formed that has uncontrolled kinase activity inside the cell. In many cases, the EML4-ALK protein promotes and maintains the malignant behaviour of the cancer cells.

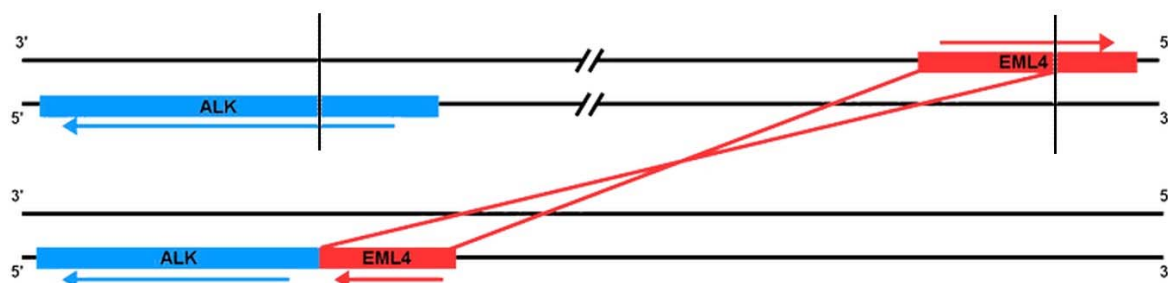


Fig. 3.1

- (a) With reference to Fig. 3.1, describe how the proto-oncogene may be converted into an oncogene. [1]

1. **Inversion of part of the *EML4* gene / chromosome, translocating/placing *EML4* adjacent to *ALK*, leading to a more active kinase/uncontrolled kinase activity;;**

Part of the study aimed to find out if normal cells are transformed to divide abnormally after treatment with the fusion gene. In the treatments, genes were introduced into normal cells that were then cultured in a flat dish containing a suitable growth medium. Abnormal cell division is indicated by the transformed cells stacking up in multiple layers called *foci*.

The four gene treatments used and the results of the cell transformation study are shown in the Table 3.1. In the fourth treatment, the *EML4-ALK* fusion gene was modified so that the kinase produced was inactive.

Table 3.1

gene treatment	formation of foci
<i>ALK</i> alone	no
<i>EML4</i> alone	no
<i>EML4-ALK</i> fusion	yes
<i>EML4-ALK</i> fusion modified	no

- (b) Explain why normal cells were observed to be in monolayer before *EML4-ALK* fusion gene was introduced. [1]

1. **Normal cells exhibit contact inhibition when in close contact with other cells / do not divide further when in contact with other cells;;**
2. **Normal cells exhibit anchorage dependence / must be attached to a substratum/physical surface;;**

(c) With reference to Table 3.1, explain how the results show that the mutation created an oncogene. [1]

1. The two genes - *EML4* and *ALK* alone do not transform cells/produce foci, only when the fusion gene is present will foci be observed;;

Note: Only fusion gene forms foci is not enough. Must refer to the two genes alone

(d) In liver cancer, *EML4–ALK* fusion gene is only present in some liver cells.

Explain why not all cells in the patient's liver are transformed into cancer cells. [3]

1. Development of cancer is a multi-step process, a single cell needs to accumulate more than one somatic mutation to become a cancerous cell;;
2. Hence, apart from inheriting genes with mutations, mutations to other genes must also occur in the somatic cell before it turns cancerous;;

(any 2 of the 3)

3. mutation of both copies of tumour suppressor gene leading to absence of suppression of cell growth and proliferation;;
4. upregulation/activation of genes coding for telomerase resulting in the cell regaining the ability to express telomerase such that the cells have limitless replicative potential/can thus divide indefinitely;;
5. mutation of genes that regulate apoptosis causing the cells to evade programmed cell death and divide indefinitely;;

[Total: 6]

- Fig. 4.1 shows his crosses and the result.

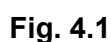


Table 4.1

(a) Draw a genetic diagram to explain the results shown in Table 4.1. [3]

Use the following symbols to represent the different alleles involved:

Gametes GN gn gn ...

	<u>GN</u>	<u>gn</u>	<u>Gn</u>	<u>gN</u>
<u>gn</u>	<u>GN</u> <u>gn</u>	<u>gn</u> <u>gn</u>	<u>Gn</u> <u>gn</u>	<u>gN</u> <u>gn</u>

Offspring genotype :	$\frac{GN}{gn}$	:	$\frac{gn}{gn}$	:	$\frac{Gn}{gn}$	:	$\frac{gN}{gn}$	::
Offspring phenotype :	grey body, long wings	:	black body, short wings	:	grey body, short wings	:	black body, long wings	
Offspring phenotypic ratio :	7 (113)	:	8 (120)	:	1 (15)	:	1 (16)	

(b) Use your knowledge of gene linkage to explain these results. [3]

1. The 2 genes for body colour and length of wings are linked and the alleles of the two genes will be inherited together as one linkage group, resulting in a higher proportion of gametes from F₁ carrying the parental types;;
2. Crossing over between the two linked genes on non-sister chromatids of homologous chromosomes may occur, as crossing over is a chance event, there are lower chances of getting recombinant gametes;;
3. thus the observed offspring of test cross are majority of parental combinations and minority of recombinants combinations;;

(c) If these genes were not linked, what ratio of phenotypes would the scientist have expected to obtain in the offspring? [1]

1. 1 grey body, long wing : 1 grey body, short wing : 1 black body, long wing : 1 black body, short wing;;

The equation to the chi-squared test and part of the critical values of the chi-squared distribution is shown below.

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad \nu = c - 1$$

Key to symbols

Σ = 'sum of'

ν = number of degrees of freedom

O = observed 'value'

c = number of classes

E = expected 'value'

df	Probability								
	0.995	0.975	0.9	0.5	0.1	0.05	0.025	0.01	0.005
1	.000	.000	0.016	0.455	2.706	3.841	5.024	6.635	7.879
2	0.010	0.051	0.211	1.386	4.605	5.991	7.378	9.210	10.597
3	0.072	0.216	0.584	2.366	6.251	7.815	9.348	11.345	12.838
4	0.207	0.484	1.064	3.357	7.779	9.488	11.143	13.277	14.860
5	0.412	0.831	1.610	4.351	9.236	11.070	12.832	15.086	16.750

- (d) Using the chi-squared distribution, apply the chi-squared test to determine whether the results obtained conformed to the expected Mendelian ratio for such a cross. [4]

Working (not required)

O	E	O-E	O-E ²	O-E ² /E	4sf
113	66	47	2209	33.4697	33.47
120	66	54	2916	44.18182	44.18
15	66	-51	2601	39.40909	39.41
16	66	-50	2500	37.87879	37.88
				$\chi^2 =$	154.94

1. χ^2 value is 155;;
2. The calculated χ^2 value is 155, more than the critical χ^2 value of 7.815 at $p = 0.05$, Value of p is less than 0.005, less than $p = 0.05$;;
3. Reject the null hypothesis, there is significant difference between the observed and the expected ratio;;
4. The results obtained do not conformed to the expected Mendelian ratio (1:1:1:1) for such a cross, any deviation from the expected is not due to chance but due to other factors eg. linkage;;

[Total: 11]

- 5 The plasma membrane of beetroot cells contains protein and phospholipids and has a porous nature. As a result, the membrane is selectively permeable.

Cyanide is a poison that inhibits cytochrome c oxidase involved in aerobic respiration.

Fig. 5.1 shows how cyanide concentration affects the uptake of chloride ions by beetroot cells. The rates of chloride ion uptake are given as percentages of those obtained in a control experiment with no cyanide.

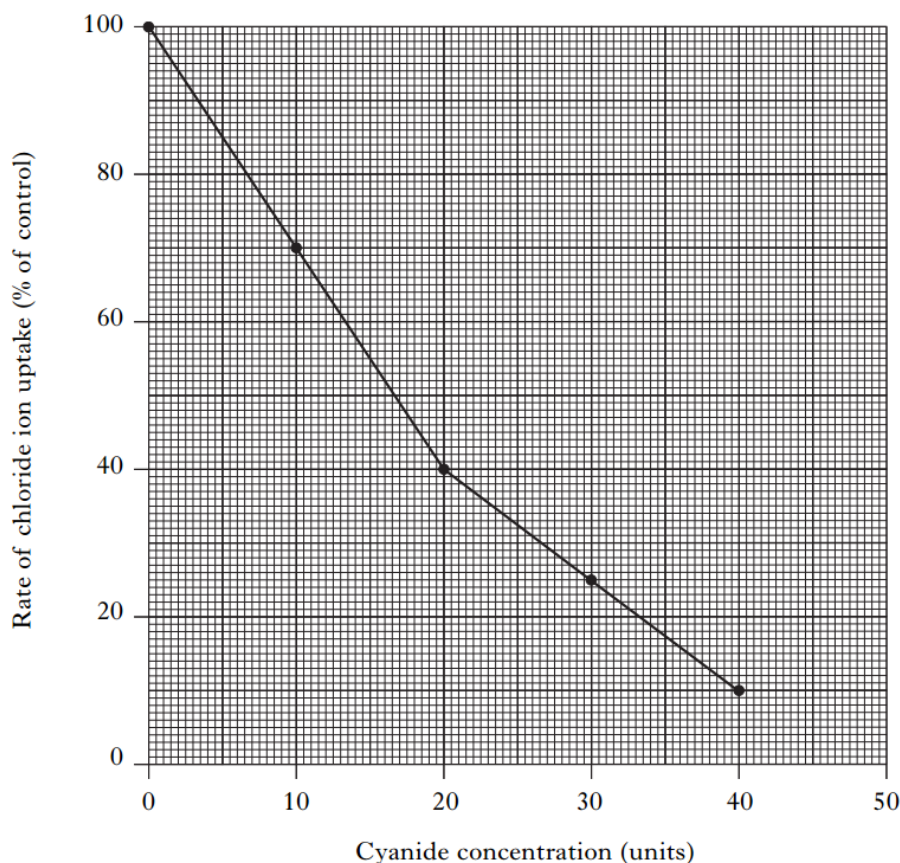


Fig. 5.1

- (a) (i) It is known that the uptake of chloride ions requires the presence of transmembrane protein.

Explain why chloride ions cannot freely pass through the phospholipid bilayer of membranes. [2]

1. **Non-polar/hydrophobic fatty acids make up the hydrophobic core of the cell membrane/phospholipid bilayer of membranes, thus the membrane is impermeable to ions;;**
2. **Cl^- are charged and cannot move across the membrane because they are repelled by the hydrophobic core of the phospholipid bilayer of membranes;;**

(b) With reference to Fig. 5.1,

(i) identify how chloride ions would be moved across the membrane, [1]

1. uptake occurs by active transport.

(ii) account for your answer in (b)(i). [3]

- 1. As cyanide concentration increases from 0 units to 40 units, rate of chloride ion uptake decreases from 100% of control to 10% of control;;**
- 2. As cyanide concentration increases, rate of aerobic respiration decrease leading to decrease in (amount of) ATP synthesized;;**
- 3. As ATP synthesis decreases, rate of chloride ion uptake decreases, indicating that ATP is required for the transport of Cl⁻;; (correlation)**

- (c) The activity of cytochrome c oxidase was measured at different values of pH by using nine different buffer solutions. The temperature was kept constant at 30°C.

The results are shown in Fig. 5.2.

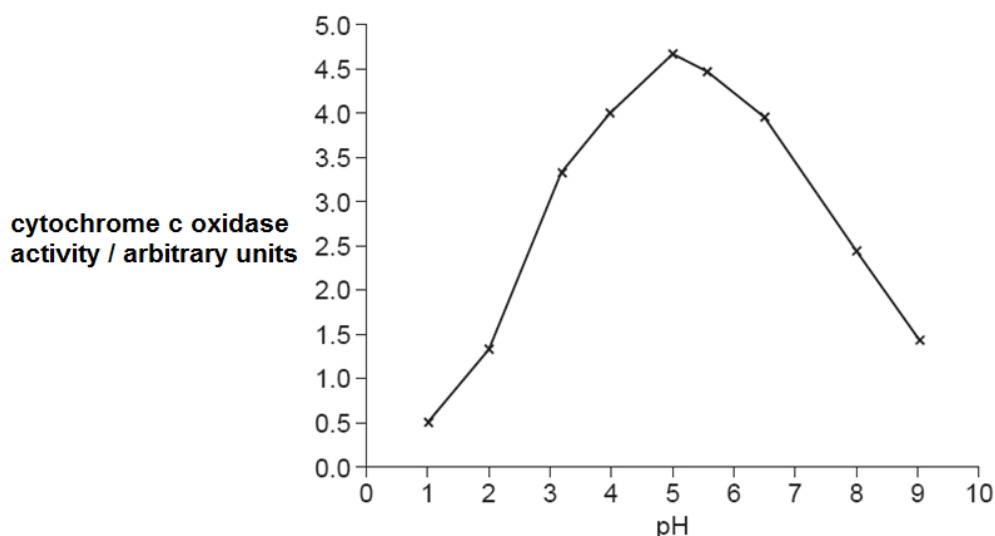


Fig. 5.2

- (i) With reference to Fig. 5.2, describe the effect of pH on the activity of cytochrome c oxidase. [2]

1. At optimum pH of pH 5, maximum activity of 5 a.u occurs;;
2. At pH other than optimum pH 5, the activity decreases, reaching 0.5 a.u at pH 1 and 1.5 a.u. at pH 9;;

- (ii) Explain why the activity of cytochrome c oxidase at pH 1 is very low. [3]

1. low pH alters the ionic charge of the acidic and basic R groups on the amino acids (at the active site of the enzyme);;
2. the ionic bonds and hydrogen bonds that help to maintain the 2° and 3° structure / specific shape of the active site of the enzyme are disrupted;;
3. causing a loss of 3D conformation of enzymes' active site, loss of 3D conformation of enzymes and the enzymes are denatured;;
4. substrate can no longer recognise and bind to the active site (of the enzyme) to form E-S complexes, decreasing the rate of formation of E-S complexes and hence decreasing the rate of formation of products/ activity;;

[Total: 11]

- 6 Fig. 6.1 shows a normal mitochondrion and a mitochondrion with a structural defect due to lethal cell injury.

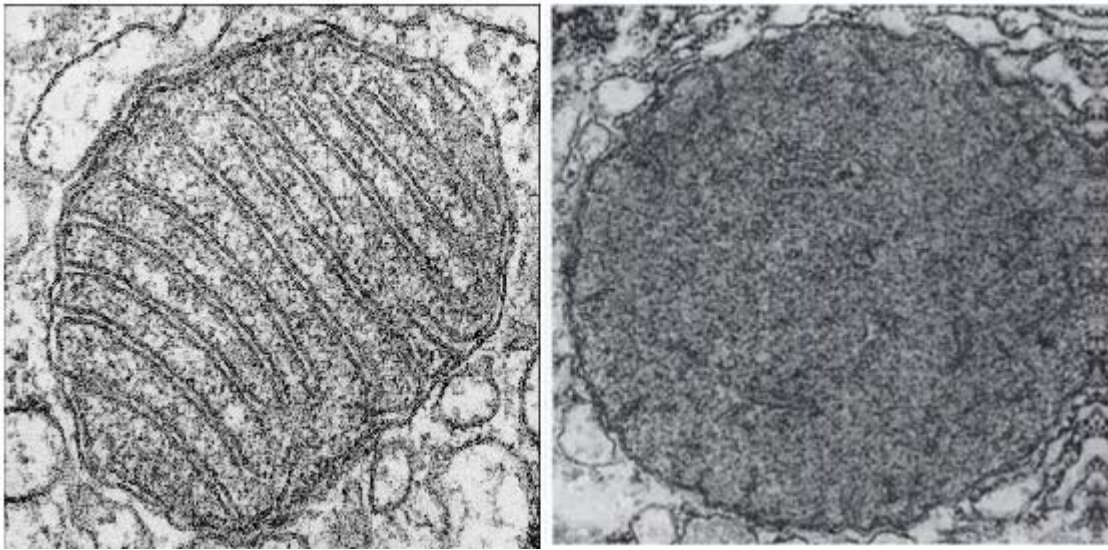


Fig. 6.1

(a) State the mitochondrial structural defect shown in Fig. 6.1. [1]

1. Cristae loss/No inner membrane/No infolding of inner membrane;;

(b) Explain the implications of this structural defect on

(i) glycolysis, [2]

- 1. Presence of cytosol, glycolysis can still occur;;**
- 2. glycolytic enzymes found in the cytosol are present;;**
- 3. fermentation occurs to regenerate NAD for glycolysis;;**

(ii) oxidative phosphorylation. [2]

- 1. Absence of cristae and therefore electron carriers of the electron transport chain (ETC) that are embedded on the cristae are absent;;**
 - 2. There will be no electron flow along the ETC and no built up of proton gradient between the intermembrane space and mitochondrial matrix, oxidative phosphorylation cannot occur;;**
- Or**
- 3. Stalked particles/ATP synthase that are embedded on the cristae are absent;;**
 - 4. ATP synthesis by chemiosmosis / diffusion of H^+ down a concentration gradient from intermembrane space to matrix does not occur and oxidative phosphorylation cannot occur;;**

- (c) The metabolic pathway in which a hexose sugar, such as glucose, is broken down in respiration by cells starts with glycolysis as shown in Fig. 6.2.

The enzymes responsible for catalyzing the first and last key steps, phosphofructokinase (PFK) and pyruvate kinase are the primary steps for allosteric enzyme regulation. PFK is allosterically inhibited by ATP, so glycolysis is slowed when cellular ATP concentrations are high.

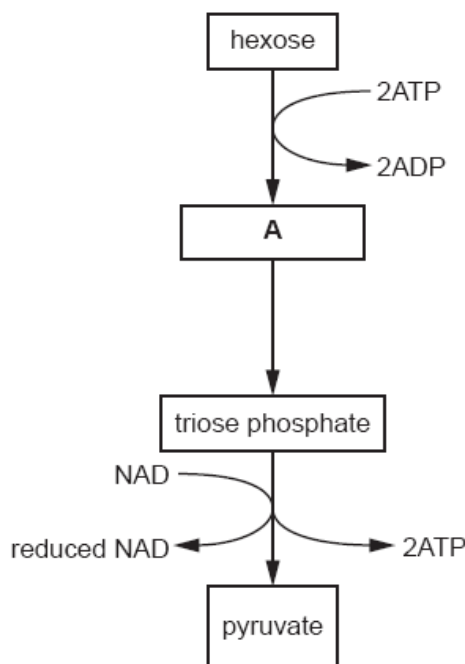


Fig. 6.2

- (i) Name substance A and explain why hexose is converted to substance A. [2]

1. **Fructose-1,6-bisphosphate;;**
2. **This is a key step for regulation of glycolysis and determines the speed (rate) at which overall glycolysis proceeds;;**
3. **With 2 phosphate groups (on its opposite ends), the sugar is now ready to be split in half;;**

- (ii) Explain the role of NAD in aerobic respiration. [2]

1. **Act as a coenzyme (for dehydrogenase) and hydrogen atom / hydrogen ion and electron carrier and is reduced to NADH during glycolysis, link reaction and Krebs cycle in aerobic respiration;;**
2. **NADH carries electrons and protons / hydrogen atoms to electron transport chain at inner mitochondrial membrane to be used in oxidative phosphorylation;;**
3. **Leading to the regeneration of NAD for subsequent glycolysis, link reaction and Krebs cycle to proceed;;**

[Total: 9]

- 7 An investigation was carried out into the effect of eating two different sources of carbohydrate on the concentration of glucose and the concentration of insulin in the blood plasma. The carbohydrate sources used were glucose and rice. The carbohydrate in rice is mainly in the form of starch.

Volunteers were fed either 50 g of glucose or the quantity of rice known to contain 50 g of carbohydrate. The concentration of glucose and insulin in their blood plasma was then measured at intervals for 3 hours. The results are shown in Fig. 7.1 and Fig. 7.2.

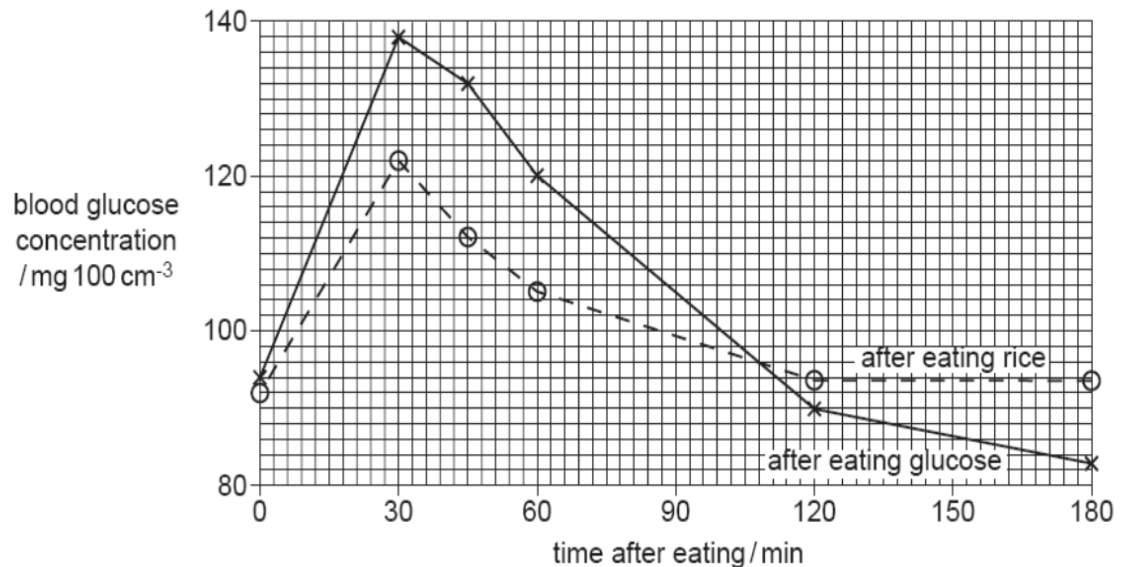


Fig. 7.1

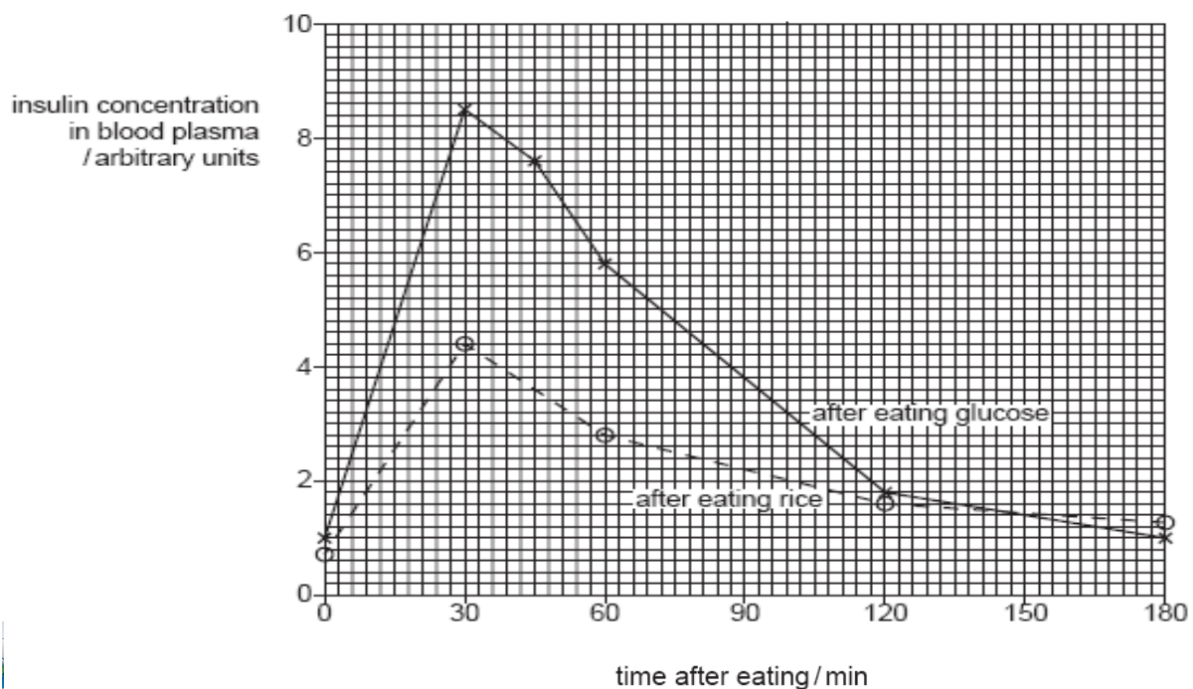


Fig. 7.2

(a) With reference to Fig. 7.1, give an explanation for the difference in concentrations of blood glucose in the first 30 minutes after eating glucose and after eating rice. [1]

1. Glucose does not need digestion but starch does so glucose is more quickly absorbed;;

(b) With reference to Fig. 7.1 and Fig. 7.2, explain the differences between the concentration of insulin in the blood at 30 minutes after eating glucose and after eating rice. [4]

- 1. Blood glucose concentration increases above normal level of 92-4mg / 100cm³ of blood;;**
- 2. Islet of Langerhans of the pancreas detect the deviation and β -cells of the islet of Langerhans of the pancreas secrete insulin;;**
- 3. Insulin concentration follows (pattern of) changes in blood glucose concentration;;**
- 4. (At 30 minutes after eating glucose and eating rice), the blood glucose concentration from eating blood glucose was 16 mg 100 cm³ more than eating rice therefore 4.1 a.u. more insulin secreted;;**
- 5. Negative feedback mechanism operates such that any deviation of the variable from the reference point triggers mechanisms to counteract that deviation and cause the variable to return to the reference point;;**

(c) By 180 minutes after eating the glucose or rice, the concentration of glucose in the blood had fallen considerably.

Describe the role of skeletal muscle in bringing about the reduction in blood glucose concentration. [1]

There will be

- 1. an increase in the permeability of plasma membrane to glucose to increase the rate of uptake of glucose from the blood by muscle cells;;**
- 2. an increase in the rate of conversion of glucose to glycogen / glycogenesis (for storage);;**
- 3. an increase in the rate of respiration using glucose as respiratory substrate (which will be broken down and oxidised to form carbon dioxide and water);;**

Fig. 7.3 shows insulin activation of a GLUT4 transporter that aids in removing excess glucose from the blood.

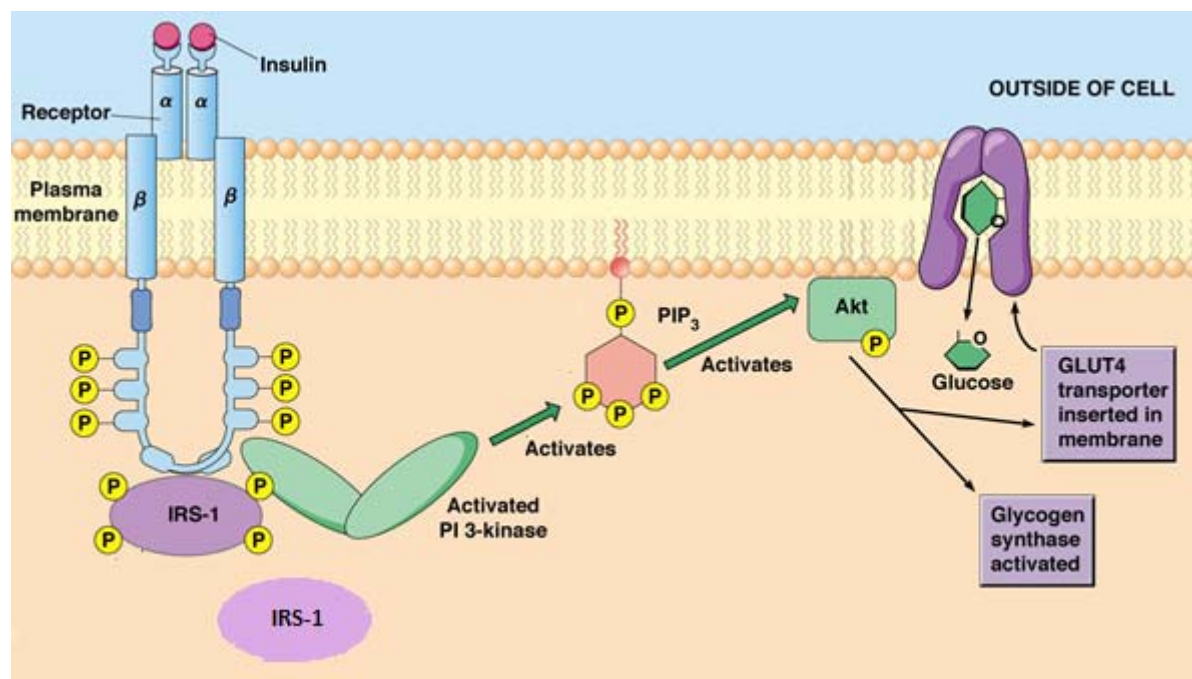


Fig. 7.3

(d) (i) With reference to Fig. 7.3, describe two advantages of this cell signalling pathway. [2]

1. Many cells can be activated simultaneously by insulin as long as the cells have the insulin receptors;;
2. One molecule of insulin can cause more than one molecule of IRS-1 to be produced for signal amplification / allow for signal amplification when some molecules (activated IRS-1 / PI-3 kinase / PIP_3 / Akt) in a pathway transmit the signal to numerous molecules (PI-3 kinase / PIP_3 / Akt) at the next step of the pathway;;
3. Allows insulin to trigger specific cellular responses due to the collection of insulin receptors, IRS-1, PI 3-kinase, PIP_3 and Akt in target cells as different kind of cells may respond differently to the same signals;;
4. Provide different points at which a cell's response can be regulated e.g. phosphorylation / activation of IRS-1 / PI 3-kinase / PIP_3 / Akt;;
5. Insulin reaching the cell membrane can trigger a series of reactions / numerous cellular reactions at once e.g. phosphorylation / activation of IRS-1 / PI 3-kinase / Akt / PIP_3 in the cytosol upon recognizing and binding to the receptor;;

(ii) Explain how GLUT4 transporter could remain embedded in the plasma membrane. [1]

1. GLUT4 transporter remains embedded as there are hydrophobic interactions between non-polar R groups of GLUT4 transporter amino acids and non-polar hydrocarbon tails of the phospholipid bilayer of the plasma membrane;;

[Total: 9]

- 8 There are over 40 Galapagos Islands including the small and isolated island named Daphne Major.

Studies were made every year from 1970 to 1989 on the beak size of the island's population of ground finch, *Geospiza fortis*, by measuring the beak length of every bird (Fig. 8.1). Larger finches with larger beaks are better at opening large seeds.



Fig. 8.1

From 1976 to 1978 there was a drought and only 15% of the ground finches survived and these did not breed during drought years. The most conspicuous feature of the survivors of the drought years was their large beak size.

During normal years, many drought intolerant grasses and herbs produce an abundance of small seeds. A few other drought-tolerant plants produce a much smaller number of large seeds which are not normally eaten.

(a) Describe how environmental factors act as forces of natural selection, during drought years, on the beak size of finches. [4]

1. **Drought-tolerant plants which produced large seeds survived the drought better;;**
2. **The selection pressure being the type / availability of food source e.g. large seeds;;**
3. **Birds with larger beaks that can feed on large seeds are at selective advantage;;**
4. **can survive to maturity, mate, reproduce and pass on their favourable alleles / genes (for larger beaks) to their offsprings;;**

(b) Suggest the role of the islands in the evolution of thirteen species of Darwin finches now found on the Galapagos Islands. [2]

1. **Geographical barrier / isolation between islands leads to reproductive isolation;;**
2. **Results in a lack of gene flow between population of finches;;**
3. **Different selection pressures on the different islands;;**

Molecular analysis was carried out on the mitochondrial DNA (mtDNA) sequences of the Galapagos islands finches and the Cocos finch found on the island of Cocos, 830 km to north-east of the Galapagos islands. Using mtDNA analysis data, a map showing the phylogeny of these finches was constructed as shown below in Fig. 8.2

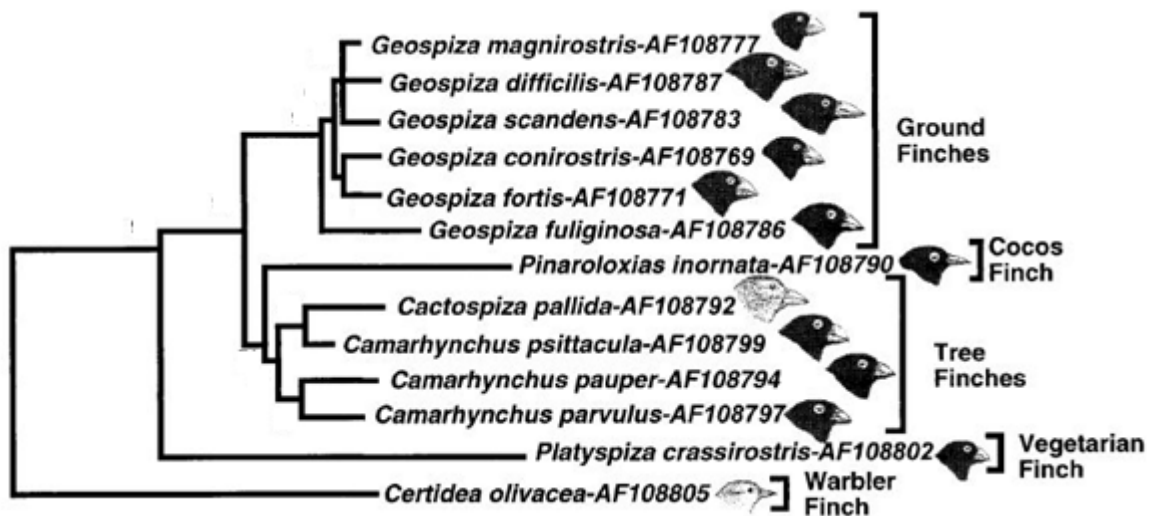


Fig. 8.2

(c) Explain how DNA sequences can be used to determine evolutionary relatedness between species. [2]

1. Compare regions of homology in the same gene e.g. cytochrome c gene / β -chain haemoglobin gene found in different species / compare homologous genes between species;;
2. The fewer the differences in DNA sequences of homologous genes between species, the more closely related the species are (vice-versa);;

(d) Describe the advantages of using nucleotide data such as mtDNA in classifying organisms. [2]

1. Using nucleotide data is unambiguous and objective. A, T, G, C are easily recognized and not confused with one another. They are not dependent on subjective judgements or observations involving quantitative differences;;
2. Nucleotide data are quantifiable and can be converted to numerical form and open to statistical and mathematical analysis. This provides a quantitative tool for constructing phylogenetic trees with branch points defined by mutations in DNA sequence;;

[Total: 10]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 9 (a)** Outline how the structure of a named structural carbohydrate is related to its specific functions. [7]

Structure: (max 4)

- 1. Cellulose molecule consists of a long (unbranched) chains of β -glucose residues joined by β -1, 4 glycosidic bond which make the chain straight;;**
- 2. Successive β -glucose residues are rotated 180° with respect to its adjacent residues;;**
- 3. which results in the $-OH$ groups projecting outwards from each chain in all directions;;**
- 4. Many cellulose chains run parallel to each other, forming hydrogen bonds between $-OH$ groups of neighbouring chains;;**
- 5. resulting in cross-linking that bind the chain rigidly together to form microfibrils and macrofibrils;;**

Function: (max 4)

- 6. Main structural component of all plants' cell wall;;**
- 7. Microfibrils confers high tensile strength, stability and support;;**
- 8. The high tensile strength of microfibrils prevents the plant cells from bursting when water enters by osmosis;;**
- 9. As a cell inflates with water, pressure develops inside it and the cell becomes turgid, which help support plants which lack wood;;**
- 10. The arrangement of fibres around the cell helps to determine the shape of the cell as it grows;;**

(b) Compare and contrast collagen and haemoglobin in terms of their structure and functions. [6]

1. made up of amino acids joined together by peptide bonds;;
2. both quaternary proteins consisting of an aggregation of two or more polypeptide chains;;

	Collagen	Haemoglobin
3. Type of protein;;	Fibrous protein	Globular protein
4. Basic unit / structure;;	Three helical polypeptide chains wound around each other, forming a tropocollagen/ triple helix	Two α chains and two β polypeptide chains
5. Bonds;;	The three polypeptide chains are held together by intermolecular hydrogen bonds	The four polypeptide chains are held together by: ionic bonds, hydrogen bonds and hydrophobic interactions
6. Formation of covalent cross-links;;	Covalent cross-links form between the carboxyl end of one tropocollagen and the amino end of another tropocollagen to form fibrils (Fibrils unite to form fibres)	No covalent cross links formed between basic unit / structure
7. Arrangement of polymers;;	Collagen fibrils run parallel to each other and arranged in a staggered manner	Polymers are not arranged in a staggered manner
8. Presence of non-polypeptide unit required for function;;	Each polypeptide chain is not covalently bound to any non-polypeptide unit	Each polypeptide chain covalently bound to a prosthetic haem group (Each haem group contains Fe^{2+} which binds to one oxygen molecule)
9. Regularity of monomer sequence;;	Almost every third residue is the same - amino acid glycine	Amino acid sequence do not show regularity
10. Function;;	Structural protein	Transport protein
11. Function;;	provide strength and structural support	found in red blood cells to transport oxygen

(c) Discuss the roles of components of telomerase and telomeres in enabling the function of stem cells. [7]

1. The active site of telomerase reverse transcriptase is complementary to/determines the specific region of DNA to be bound and hence the region of DNA to be elongated;;
2. catalyses the elongation of DNA by adding telomere repeat sequences (TTAGGG) to the 3' end of DNA strand;;
3. via complementary base pairing of new deoxyribonucleotides using telomerase RNA as template with the formation of phosphodiester bonds between adjacent deoxyribonucleotides;;
4. telomerase RNA contains the nucleotide sequence (- 3' AAUCCC 5') that forms complementary base pairs with the 3' end of the DNA;;
5. provides a template for the complementary base pairing of new deoxyribonucleotides at the 3' end of the DNA;;
6. allows the telomere sequences to be repeated and extend the telomere;;
7. Telomeric DNA are bounded by specific proteins which protect the chromosomal ends from degradation by cellular exonucleases.
8. Telomeres protect and stabilize the terminal ends of chromosomes by forming a loop. This prevents accidental fusion of the single-stranded end of one chromosome to the single stranded end of another chromosome via complementary base pairing.
9. Telomeres prevent loss of genes/coding sequences as the DNA shortens with each round of DNA replication due to the inability of DNA polymerase to replicate to the end of the chromosome and complete the 5' ends of newly synthesized DNA strands.
10. Telomeres that are critically short trigger apoptosis (programmed cell death) to limit number of cell division.

[Total: 20]

10 (a) Describe the behaviour of chromosomes during the mitotic cell cycle. [8]

Prophase:

1. Chromatin shorten and thicken/condense and become distinct structures called chromosomes;;

Metaphase:

2. Spindle fibres from each pole of the cell are attached to one of the two chromatids of each chromosome at the centromere region;;
3. Spindle fibres pull on the centromere, arranging the chromosomes in a single row on the metaphase plate (equator of the spindle);;

Anaphase:

4. Centromere divides and sister chromatids separate at centromere, forming daughter chromosomes;;
5. Shortening of microtubules occurs and spindle fibres attached to the centromeres pull the daughter chromosome to the opposite poles of the cell;;
6. with centromere leading the way;;

Telophase:

7. Daughter chromosomes reach the opposite poles;;
8. Chromosomes uncoil, lengthen and become indistinct to form chromatin again;;

2 correctly named stages – 1 mark

(b) Explain the importance of mitosis in growth, repair and asexual reproduction. [6]

1. Mitosis produce daughter cells which are *exact copies* of the parent cell, forming genetically identical cells;;

Repair

2. Damaged and worn out cells must be replaced by exact copies of the originals to return a tissue to its former condition;;
3. e.g. cells in the skin that die from the normal wear and tear;;
4. Mitosis enables some animals to regenerate whole / missing parts of the body;;
5. e.g. tails of house lizard, legs in crustacean and arms in starfish;;

Growth

6. Mitosis enables a multicellular organism (such as a human) to grow and develop from one single cell (the fertilised egg);;
7. Whole organisms grow by increasing in cell numbers by the process of mitosis before differentiation of cells into specialised cell types occurs by switching certain genes on and off;;
8. Mitosis results in new cells which are identical to existing cells and enables growth and enlargement of tissue;;

Asexual Reproduction

9. Mitosis result in genetic stability within populations of cells derived from the same parental cell e.g. in a clone;;
10. Mitosis enables certain animals and plant species to be propagated by asexual reproduction resulting in clones so that the species can colonise a particular stable, well-adapted habitat within the shortest period of time;;

(c) Describe the differences between continuous and discontinuous variation. [6]

	Features	Continuous Variation	Discontinuous Variation
1.	Types of classes	Continuous variation giving a range of phenotypes / Characteristics fall within a range of intermediate values between 2 extremes;	Discontinuous variation giving discrete phenotypic classes / Characteristics fall into distinct classes with no intermediates;
2.	Number of genes involved	It is controlled by the combined effect of a number of genes/ polygenes;	It is controlled by 1 or 2 major genes/ few genes;
3.	Effects of genes	Each gene has a small additive effect on the character / phenotype/phenotypic expression is the result of additive/combined effect of these genes/ alleles;	Effects of the genes are absolute/ no additive effect / Different alleles at a single gene locus have large effects and different gene loci have quite different effects on the character;
4.	Effects of environment	The environment has a large effect on the phenotype / greatly affected by the environment.	The environment has a small or no effect on the appearance of the phenotype / not greatly affected / largely unaffected by the environment.
5.	Observation of effect of gene	Effect of individual polygenes cannot be observed. Effect of polygenes is additive	Effect of individual genes can be observed
6.	Type of inheritance/ Characteristics;	Inheritance/ Characteristics is quantitative.	Inheritance/ Characteristics is qualitative.
7.	Method of measurement;	Mechanisms of inheritance investigated using statistical methods / Traits/ characteristics measured quantitatively by measuring (characters of degree).	Mechanisms of inheritance investigated by counting and comparing ratios in the offspring / Traits/ characteristics measured qualitatively by counting (characters of kind).
8.	Examples;	In humans: Height/ weight	In humans : sex/ blood group

[Total: 20]